SignFlow complete project documentation

Case Study no. 89 — SignFlow: Digital Document Approval Workflow for a University
Course: Software Engineering & Project Management, Semester III
Batch: B.Tech CSE 2025–29

This package is a documentation-only submission. It does not claim to contain a fully functional application or a deployed service. It contains the business case, requirements, process/design diagrams, project planning, estimation, testing, risk management, and static UI representations needed to explain the proposed system.

Deliverables

1. `01_SRS.md` — IEEE 830-style Software Requirements Specification.
2. `02_BRD.md` — Business Requirements Document with problem, objectives, scope, stakeholders, business rules, options, and acceptance.
3. `03_UML_Package.md` — Use-case, class, sequence, activity, and state diagrams with cohesion/coupling justification.
4. `04_Project_Plan.md` — WBS, dependency network, critical path, schedule, milestones, resources, and monitoring/control.
5. `05_Estimation_Sheet.md` — Step-by-step effort calculations for the all-types and top-three options.
6. `06_Test_Plan_and_Evidence.md` — Test plan, BVA, equivalence classes, routing decision table, traceability, defect evidence, DRE, and defect density.
7. `07_Risk_Register_and_Closure_Note.md` — Risk register, RMMM, issue log, closure conditions, handover, and lessons learned.
8. `08_UI_Design_Specification.md` — Information architecture, screen inventory, interaction rules, accessibility notes, and UI acceptance checklist.
9. `09_Complete_Project_Documentation.md` — Submission index and how the documents fit together.
10. `uml/` — Editable Mermaid sources and rendered diagrams, including architecture and Gantt chart.
11. `ui/` — Five redesigned PNG UI screens, with editable HTML/CSS sources under `ui/source/`: dashboard, new file, inbox, file detail, and reports.
12. `pdf/` — PDF versions of all Markdown documents.

Key supplied figures

- Monthly volume: 2,000 files.
- Purchase requests: 700; leave: 600; event approvals: 300; circulars: 150; others: 250.
- Workflow design: 3 days per file type; configuration: 4 days; testing: 2 days.
- File types in scope for the all-types option: 12.
- Target average approval time: under 4 days.
- Target lost files: zero.

Important assumptions

Where the case study did not provide exact calendar dates, staff capacity, signature provider, amount thresholds, input limits, or executed defect counts, the documents label the value as an assumption, placeholder, planning value, or illustrative evidence. Those values should be replaced during requirements validation rather than treated as project facts.